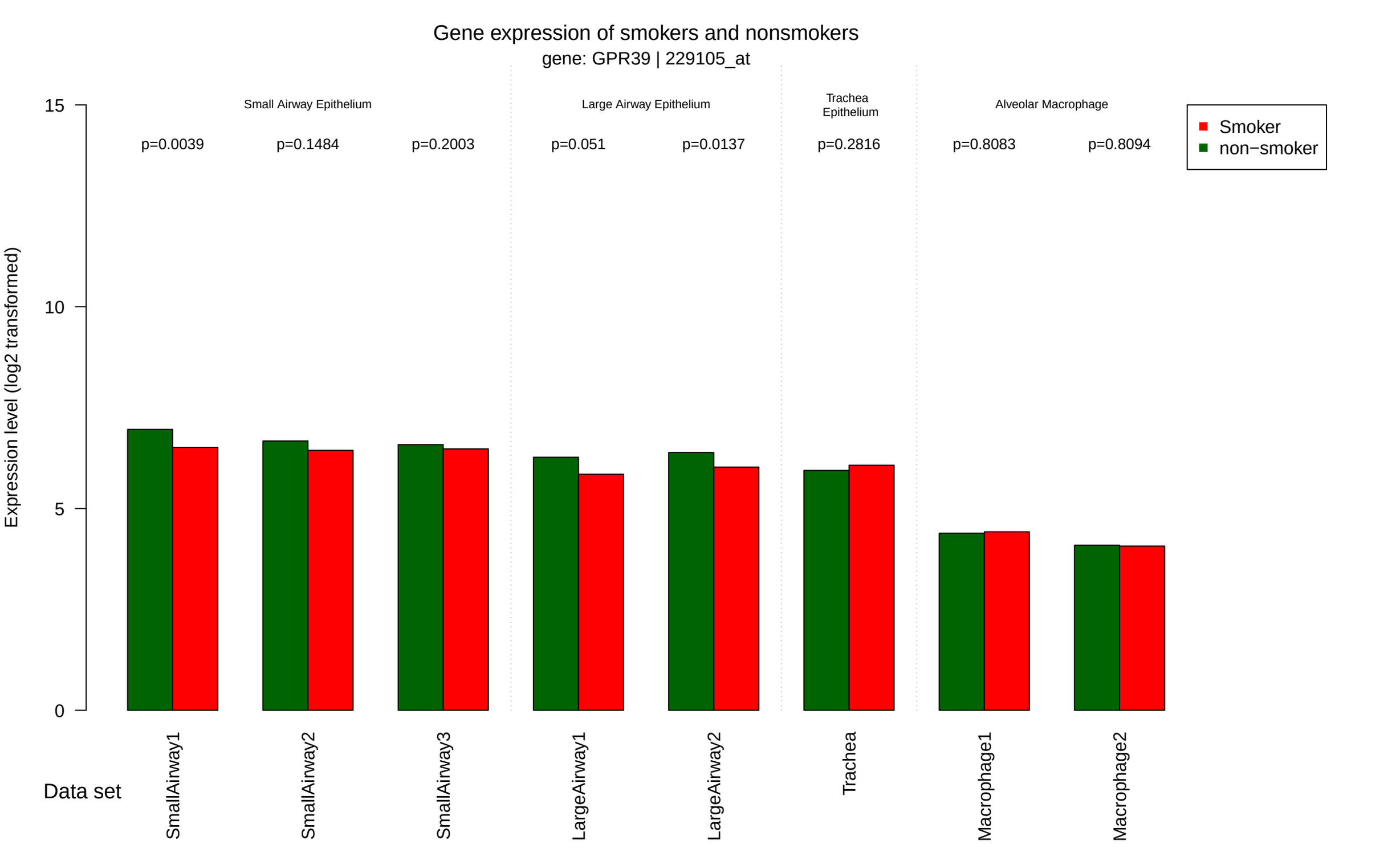
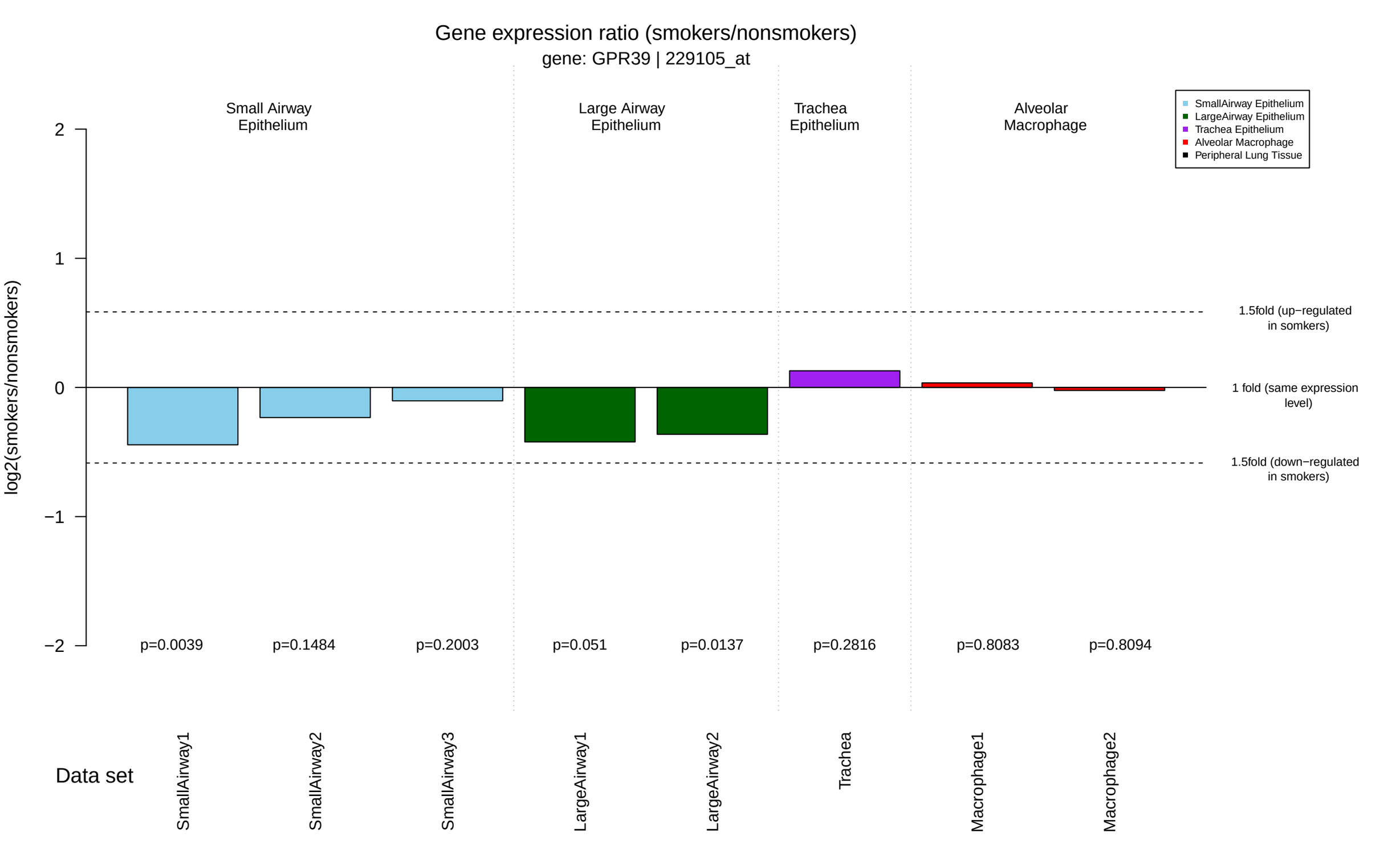
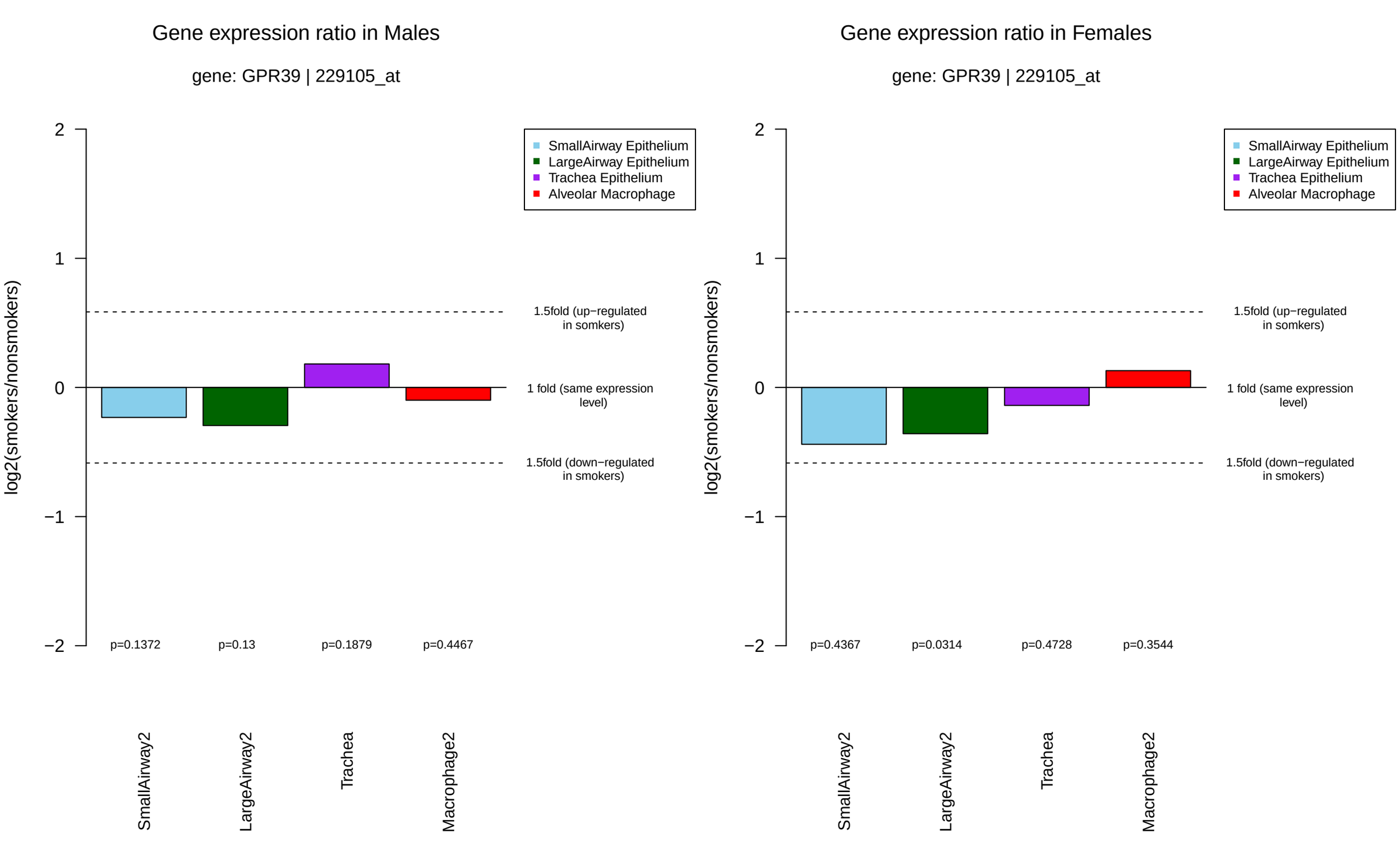
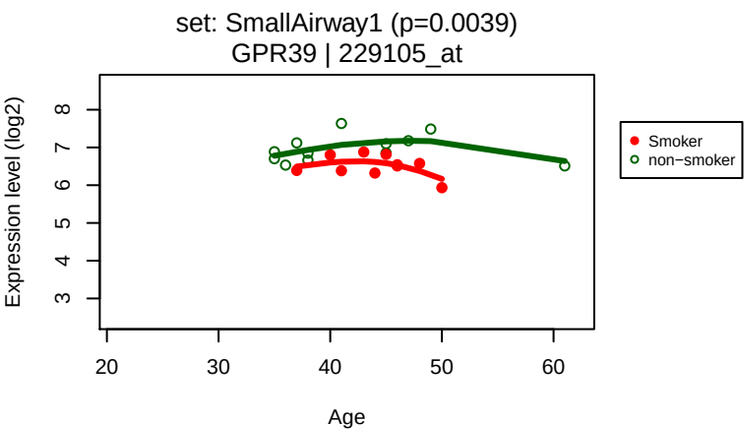
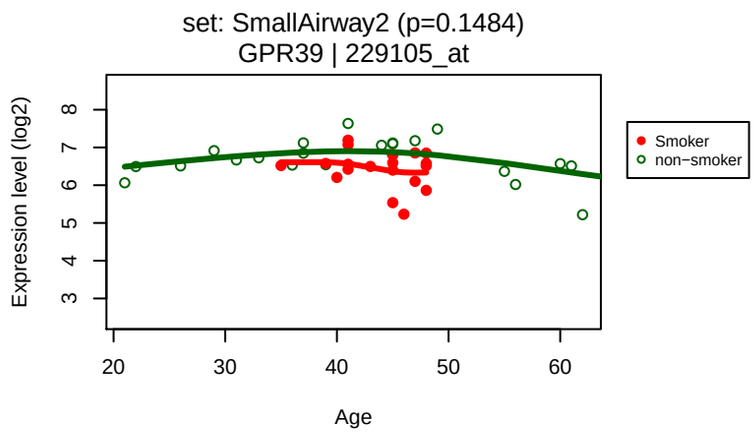
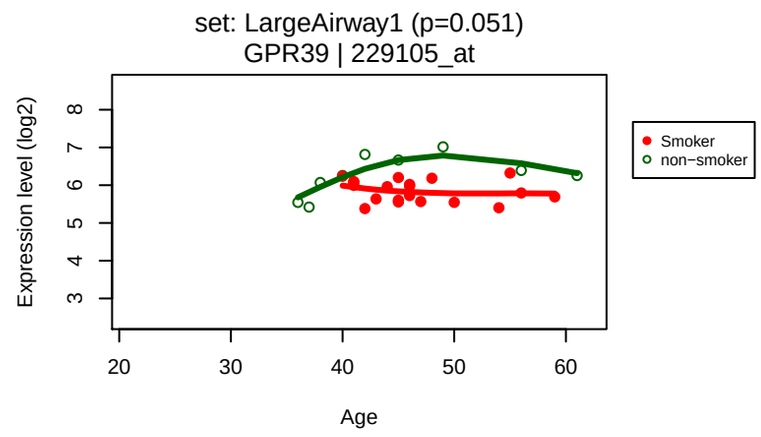

Gene expression of smokers and nonsmokers
gene: GPR39 | 229105_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0039
p=0.1484
p=0.2003
p=0.051
p=0.0137
p=0.2816
p=0.8083
p=0.8094
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

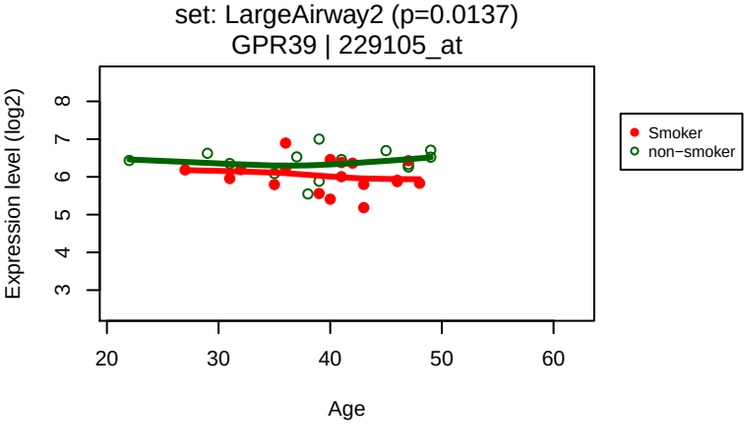

Gene expression ratio (smokers/nonsmokers)
gene: GPR39 | 229105_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0039
p=0.1484
p=0.2003
p=0.051
p=0.0137
p=0.2816
p=0.8083
p=0.8094
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

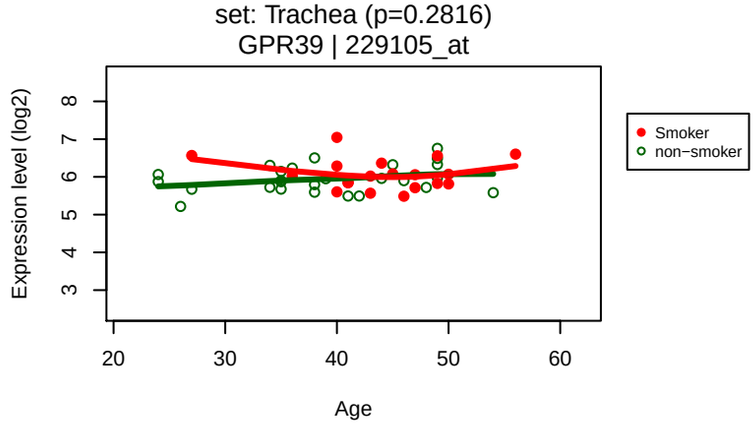

Gene expression ratio in Males
gene: GPR39 | 229105_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1372
p=0.13
p=0.1879
p=0.4467
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR39 | 229105_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4367
p=0.0314
p=0.4728
p=0.3544
SmallAirway2
LargeAirway2
Trachea
Macrophage2

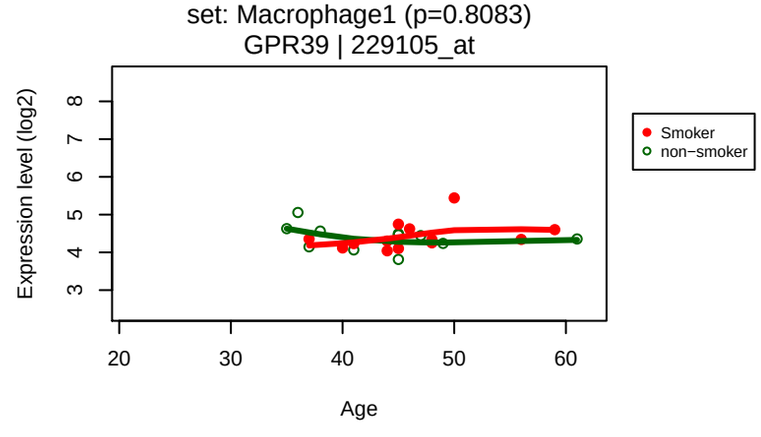

set: SmallAirway1 (p=0.0039)
GPR39 | 229105_at
Expression level (log2)
Age
Smoker
non−smoker

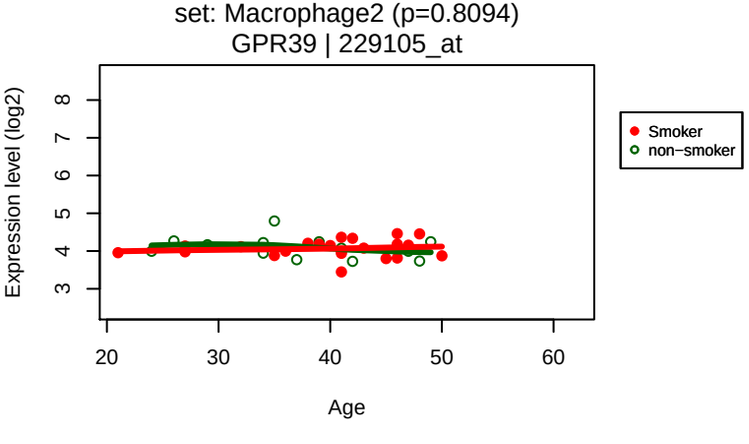

set: SmallAirway2 (p=0.1484)
GPR39 | 229105_at
Expression level (log2)
Age
Smoker
non−smoker

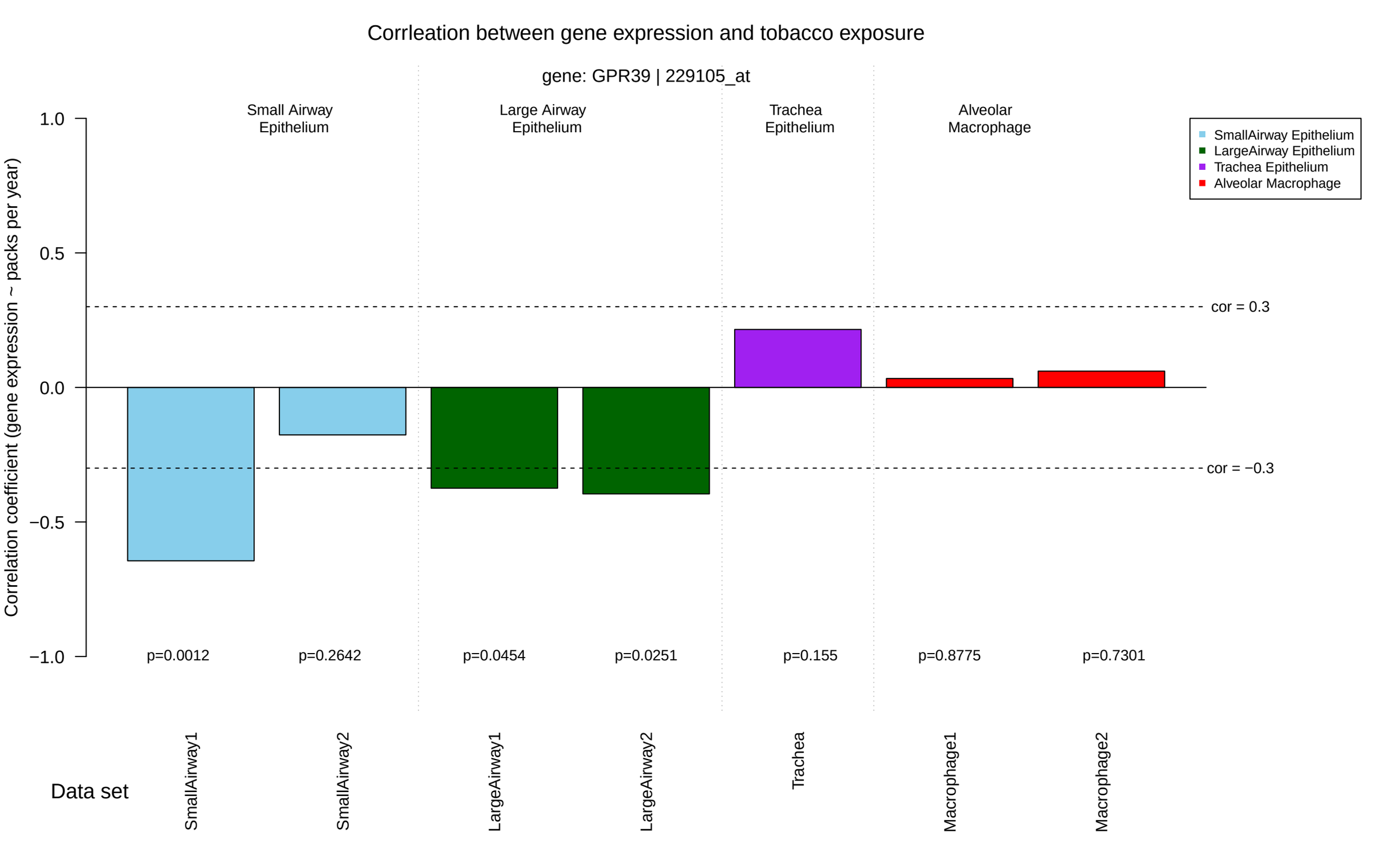

set: LargeAirway1 (p=0.051)
GPR39 | 229105_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0137)
GPR39 | 229105_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2816)
GPR39 | 229105_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8083)
GPR39 | 229105_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8094)
GPR39 | 229105_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR39 | 229105_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0012
p=0.2642
p=0.0454
p=0.0251
p=0.155
p=0.8775
p=0.7301
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2